

VIKSIT BHARAT YOUNG LEADERS DIALOGUE

HACK FOR SOCIAL CAUSE

# AASHRAY

AI-Powered Disaster Intelligence and Human-Safety Network

FULL RESEARCH · PRODUCT DESIGN · TECHNICAL ARCHITECTURE

Judge evaluation document for repository submission

Problem, thesis, features, GIS, AI, architecture, stack, workflow

Prototype scenario: landslide + rainfall + road blockage + isolation

Three-person student team · September 2026

*AASHRAY reduces uncertainty during disasters by fusing fragmented citizen, geographic and environmental signals into an explainable operational picture — and by explicitly showing responders where that picture is still incomplete.*

Prototype research paper · Not a substitute for official disaster-management systems

# Contents

Abstract

1. Introduction and national context
  2. Problem statement
  3. Research thesis and contributions
  4. What AASHRAY is not
  5. Users, operating loop, and glossary
  6. Related approaches and why they fail the mission
  7. Product design and final feature set
  8. Critical evaluation (innovation, risks, ethics)
  9. Prototype scenario and demo narrative
  10. System architecture
  11. Technology strategy (definitions, alternatives, trade-offs)
  12. Data model and APIs
  13. Incident fusion, AI, and decision engine
  14. GIS, routing, isolation, alerts, visibility
  15. Complete data flow and tick-by-tick workflow
  16. Privacy, misinformation, and failure modes
  17. Real versus simulated data
  18. Team roles and implementation plan
  19. Evaluation criteria and judge questions
  20. Limitations, scalability, and future work
  21. Conclusion
- Appendix A. Final stack table
- Appendix B. Claims we will not make

## Abstract

During floods, landslides, cyclones, fires, earthquakes and major accidents, responders rarely lack messages. They lack a single coherent picture: what is one event versus many, who has confirmed safety, which roads still work, which settlements may be isolated, and where the operational picture is blind. Generic SOS applications send coordinates. They do not fuse fragments, they do not treat uncertainty as a first-class field, and they optimise travel time rather than safety.

**AASHRAY** is an AI-assisted disaster intelligence and human-safety network designed for the Viksit Bharat Young Leaders Dialogue (VBYLD) Hack for Social Cause. It combines citizen evidence (SOS, reports, optional media, safety check-ins) with GIS (OpenStreetMap roads and facilities, terrain-informed zones, candidate routes) and deterministic operational rules (fusion thresholds, corroboration, explainable priority, access-cut detection). A large language model is used only to interpret short text and draft sitreps. Geography, reachability and ranking never depend on it.

This paper is the complete research and technical design for repository evaluation. It specifies the product thesis, a judged prototype scenario (landslide + rainfall + blockage + isolation), the software architecture for a three-person student team, every major technology with alternatives rejected, end-to-end data flow, privacy constraints, and an honest boundary between live computation and labelled simulated layers. The intended contribution is not a national command system. It is a credible, explainable *situation-awareness pipeline* that still runs if the model API is down.

*Thesis: AASHRAY is a disaster situation-awareness system that fuses citizen reports, safety status and GIS context into a live, explainable operational picture — and explicitly surfaces where that picture is incomplete.*

## 1. Introduction and national context

India's disaster burden is geographic and social: Himalayan and Western Ghat landslides after monsoon pulses, flooded plains, cyclonic coasts, urban fires, and infrastructure collapse. Official systems (NDMA/SDMA, SDRF/NDRF, 112, district control rooms) remain the lawful backbone. Citizen communication, however, fragments across calls, WhatsApp groups, unverified social posts and incomplete weather context. The first hours are an information problem as much as a logistics problem.

AASHRAY is a student prototype that asks: can we reduce that uncertainty with tools a three-person team can actually demonstrate — OpenStreetMap, a simple backend, transparent scores, and a time-evolving hill-district scenario — without pretending to replace government? The answer this paper defends is yes, if the operational object is a **fused incident with confidence**, if **visibility gaps** are mapped honestly, and if routes are chosen for **safety rather than shortest time**.

The prototype is scoped to one disaster type done deeply. Multi-hazard national coverage, live IMD coupling, and departmental dispatch are future work and are labelled as such. AASHRAY is submitted as research-backed software design plus a demonstrable workflow, not as certified life-safety equipment.

## 2. Problem statement

When a hillside fails, four people may independently report: “landslide,” “road blocked,” “vehicle trapped,” and a photograph of debris. Weather may already show intense rain. A downstream hamlet may have a single access road. Some residents may open an app and mark themselves safe; many will not. A control room that treats each message as a separate emergency wastes teams. A control room that treats silence as “missing persons” commits an ethical and operational error. A navigation app that sends ambulances along the fastest polyline through the debris field commits a geographic error.

### Operational questions AASHRAY must support

- **What is happening?** Consolidate reports into probable events.
- **Where is it happening?** GIS location, not a chat transcript.
- **Who needs help?** Structured SOS with people-count, trapped/injury flags, severity.

- **Who is safe?** Opt-in check-ins among registered scenario users.
- **Whose status is unconfirmed?** Visibility of non-check-in — never equated with missing.
- **What could happen next?** Rule-based one-hop cascade (blocked last road → isolation → delayed medical access), labelled as estimation.
- **How should responders act?** Explainable priority, nearest suitable accessible resource, safest remaining route, targeted alerts.

Central problem in one sentence: **responders do not have a sufficiently clear, live picture of events, people, routes and uncertainty**. AASHRAY attempts to reduce that uncertainty. It does not claim to eliminate it.

### 3. Research thesis and contributions

#### Thesis

AASHRAY reduces uncertainty during disasters by fusing fragmented human, geographic and environmental signals into actionable intelligence, while displaying confidence, provenance and visibility gaps. A stronger formulation used throughout this paper: the product is **situation awareness with explicit incompleteness**, not disaster prediction.

#### Contributions (what is technically new in a student prototype)

- **Evidence versus incident:** raw SOS/reports are evidence; fused clusters are the operational objects.
- **Explainable fusion confidence:** weighted geo, time, category and optional text similarity — not a mysterious accuracy percentage.
- **Human Safety Visibility Score:** where the picture is weak among simulated registered users.
- **Access-cut detection:** road-graph reachability after blockage (isolation), not pin proximity.
- **Safest-of-N routing:** score candidate OSM paths against hazard polygons; reject the fast unsafe path.
- **Terrain-clipped geofencing with corroboration:** Critical alerts are not a kilometre radius around one panic tap.
- **AI isolation:** language models may classify text and draft sitreps; they may not move roads, ranks or geometry.

#### SENSE → UNDERSTAND → PRIORITIZE → ACT → VERIFY

SENSE: citizen + environmental + GIS signals. UNDERSTAND: fusion and structured interpretation. PRIORITIZE: who/what first, with visible weights. ACT: targeted alert, safest route, resource suggestion, simulated assignment. VERIFY: check-ins and responder status. Repeat on every new evidence row or demo tick.

### 4. What AASHRAY is not

- Not another SOS button that only forwards GPS.
- Not a Google Maps clone.
- Not an official NDMA/SDRF/112 system and not a live government alerter in this prototype.
- Not a scientific landslide nowcast (“this slope WILL fail”).
- Not a missing-persons registry. Unconfirmed safety status ≠ missing.
- Not a mesh-network radio layer.
- Not a multi-hazard national digital public infrastructure in version one.

### 5. Users, operating loop, and glossary

#### Primary users

| Role                              | Can do in the prototype                                                                                                                                              | Must not see                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Citizen                           | Report emergency; SOS with structured fields; optional photo; safety status (safe / need assistance / in danger); view own pin, alert zones, safest route, shelters. | Other people's exact SOS, victim photographs, full ops action queue.          |
| Responder (simulated ops console) | Live fused incidents, evidence locker, severity, zones, safety aggregates, visibility grid, blocked roads, POIs, priority, resources, assign/dismiss.                | A fake "message sent to NDMA" unless a real integration exists (it does not). |

Table 1. Two surfaces only. A third "ministry portal" without authority is theatre.

## Glossary

| Term                      | Meaning in AASHRAY                                                                    |
|---------------------------|---------------------------------------------------------------------------------------|
| Evidence                  | One signal: SOS, report, photo metadata, or check-in.                                 |
| Incident                  | Probable real-world event after fusion; has members, geometry, confidence.            |
| Cluster confidence        | Fusion score 0–1 shown as %. Not field-validated accuracy.                            |
| Decision engine           | Deterministic GIS + rules: zones, visibility, isolation, priority, resources, routes. |
| Visibility                | How much safety status we have in a grid cell. Not a casualty count.                  |
| Unconfirmed safety status | Registered scenario users who have not checked in.                                    |
| Access-cut / isolation    | Settlement unreachable from a hub after blocked edges are removed.                    |
| Safest-of-N               | Among a few candidate polylines, prefer lowest hazard intersection, then time.        |
| Corroboration gate        | Critical-zone alerts need more than a single unverified SOS.                          |
| Demo clock                | Discrete ticks injecting rain layers and scripted evidence through the same APIs.     |
| Provenance                | Every map feature: live   fused   rule   simulated, plus as-of time.                  |
| Sitrep                    | Short summary of already computed state. A view, not an authority.                    |

Table 2. Shared language for UI copy and evaluation.

## 6. Related approaches and why they fail the mission

| Approach                                  | What it does                                          | Why it is not AASHRAY                                                          |
|-------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------|
| Consumer SOS / safety apps                | Share live location with contacts or emergency number | No fusion, no visibility of silence, no safety-first routing, no access-cut.   |
| Google/Apple Maps                         | Shortest or typical-time paths                        | Optimises time; does not know your debris polygon or last remaining hill road. |
| Social media walls                        | High volume citizen video                             | Duplicates, rumour, no operational object, no graph.                           |
| Generic "AI chatbot + map" hackathon apps | LLM describes a pin                                   | Language is not geography. Un-auditable ranks fail a district officer.         |
| National GIS dashboards (conceptual)      | Authoritative layers                                  | Out of student SLA; AASHRAY does not claim to be NDMA Sahana/SACHET.           |

Table 3. We compete with fragmentation and uncertainty, not with 112.

## 7. Product design and final feature set

An initial twelve-feature wishlist was reduced so a three-person team can demonstrate one scenario extremely well. Cascade, next-affected zone and the risk layer became one Risk and Propagation panel. Resource matching and priority became one Prioritized Action Queue. Alerts and geofences are the same model. Chatbots, mesh networking and national multi-hazard coverage were removed from the headline.

### MUST HAVE (prototype)

- Structured SOS/report as **evidence** (GPS, type, time, people, severity, trapped/injury, optional photo).
- **Incident fusion** with confidence and an evidence locker.
- **GIS operations map**: incidents, scenario rain/landslide risk, OSM hospitals/fire/shelters, check-ins, blocked roads, alert zones.
- I'm Safe / Need assistance / In danger, with **Unconfirmed Safety Status** wording and simulated registered population.
- **Human Safety Visibility** on a committed grid.
- Terrain-informed **Critical / Warning / Nearby** zones; corroboration before Critical notify.
- **Explainable Response Priority** plus nearest **suitable accessible** resource (availability may be simulated).
- **Safest available route** among 2–3 candidates; reject those intersecting hazard/blocked polygons.
- **Isolation / access-cut** when the last road to a settlement is blocked; one-hop cascade flag.
- **Demo clock** from quiet map → rain → reports → fusion → zones → visibility → reroute → isolation.

## SHOULD HAVE

- LLM responder sitrep from fused JSON.
- Optional vision tags on photos, always “possible / low confidence.”
- Offline last-snapshot cache, queued SOS, SMS-shaped payload card (no DLT gateway required).
- Conflicting-report state on a cluster.

## FUTURE

Household/ward check-in; official agency login; real IMD/CWC with fallback still labelled; PostGIS at scale; vernacular voice; production SMS; multi-hazard scientific models. None of these are required to evaluate the thesis.

## GIS layers — each must change a decision

| Layer                      | Decision                                                 |
|----------------------------|----------------------------------------------------------|
| Incident                   | What is one event; where to look first.                  |
| Risk (rain/slope scenario) | Where not to send people; which routes to penalize.      |
| Infrastructure             | Which hospital/fire/shelter/road is usable and suitable. |
| Human safety               | Declared safe / assist / danger.                         |
| Visibility / unconfirmed   | Where to send information-gathering, not a missing list. |
| Evacuation                 | Safest remaining path; blocked and dangerous edges.      |
| Alert zones                | Who is notified and at what intensity.                   |

Table 4. A decorative map fails this table.

## 8. Critical evaluation (innovation, risks, ethics)

### What is genuinely innovative here

Fusion-with-evidence, visibility-as-a-layer, and access-cut on a hill road graph. Together they answer a question SOS apps never ask: *where are we flying blind, and is the village still connected?*

### What was generic and removed

Standalone coordinate SOS, nearby-hospital lists without access context, disaster chatbots, fake government SMS, mesh networking as a poster feature, and ten half-built hazard types.

### Credibility traps we refuse

- “Expected population 3000” is **simulated registered users in the scenario**, never census.

- Cluster confidence is not “91% accurate in the field.”
- Unconfirmed is not missing.
- Rain in the demo is a labelled scenario layer unless a live API is actually wired.

False SOS, spam, misinformation

Ingest is rate-limited. Single-source reports stay unverified. Critical geofence alerts require corroboration (second report or SOS plus high scenario rain). Fusion records disagreement instead of amplifying the first viral phrase. Responders can dismiss suspected hoax. Nearby alerts are quieter (“avoid travelling toward”). One alert per user per incident per zone level until the level changes.

9. Prototype scenario and demo narrative

Recommended setting: a hill district (Himachal / Uttarakhand / Western Ghats language). Demo title: **From noise to a picture — including what we still don’t know.**

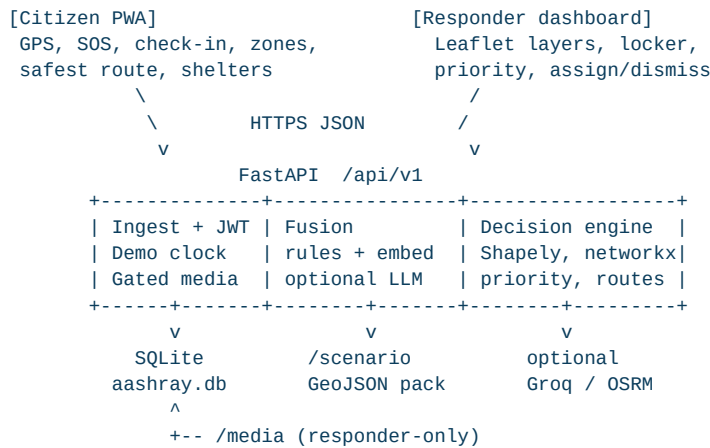
| Beat     | Jury should see                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------|
| T0       | Quiet map; rain low; visibility high in town, thin in a hamlet.                                         |
| T1       | Rain intensifies. Risk polygons rise. No panic Critical alerts.                                         |
| T2       | Mismatched reports: landslide, blocked road, trapped vehicle, debris photo. Raw feed looks chaotic.     |
| T3       | Fusion: one cluster ~85–92% cluster confidence; evidence locker. Not four pins.                         |
| T4       | Zones follow slope/valley, not a perfect circle. Nearby/Warning; Critical gated.                        |
| T5       | Town fills green; hamlet stays unconfirmed / low visibility — copy: not missing.                        |
| T6       | Fast route crosses hazard and is rejected; slower safer route recommended; hospital by type and access. |
| T7 (wow) | Last access edge blocked → village isolated; medical access delayed; priority weights jump on screen.   |
| T8       | Responder assigns a team; citizen sees status; optional 20-word sitrep from JSON.                       |

Table 5. Success metric: this clock in about three minutes, with LLM down still completing T3–T7.

*Jury line: we do not claim to know who is missing. We claim to show where the picture is strong, where it is fused, and where responders are flying blind — in time to choose a safer road and a first village.*

10. System architecture

Greenfield application **aashray/**. AASHRAY is not a module inside unrelated products. One frontend (two route trees), one FastAPI process (ingest + fusion + GIS rules + demo clock), one SQLite file, one committed scenario GeoJSON pack. No microservices, no message bus, no second “model” server — a split process is a jury-Wi-Fi failure mode.



Transport is REST JSON. The ops UI polls **GET /api/v1/ops/snapshot** about every two seconds. WebSockets look more “live” but fail more often on student hosting. Discrete demo ticks do not need a socket.

## 11. Technology strategy (definitions, alternatives, trade-offs)

Selection rule: technically credible, demonstrable by three students, reliable on demo day, inexpensive, integrable, and scalable enough for the *concept* (district-demo now; PostGIS later). Advanced-sounding infrastructure that cannot be operated in a college lab is rejected.

### 11.1 Classification key

- **AI** — optional LLM classify/sitrep, embeddings, optional vision tags.
- **GIS** — Leaflet, GeoJSON, Shapely, OSM extracts, route geometries, road graph.
- **Deterministic rules** — fusion threshold, corroboration, priority formula, isolation BFS, alert membership.
- **Database** — SQLite rows.
- **External API** — optional LLM, optional OSRM, OSM tiles.
- **Simulated data** — rainfall, population denominator, ambulance availability, extra scripted reports.
- **Live data** — optional team-phone GPS/SOS during demo; otherwise clock-injected evidence labelled simulated.

### 11.2 React 19 + Vite + TypeScript + Tailwind CSS 4

Classification: Frontend

- **Definition:** A single-page application toolchain: components, fast bundling, typed contracts, utility-first CSS.
- **Why AASHRAY uses it:** AASHRAY has two dense UIs (mobile citizen capture and desktop ops map) that share types (Evidence, Incident, Snapshot).
- **Where it is used:** aashray/frontend — routes `/c/*` and `/ops/*`.
- **Why not the alternative:** Next.js adds server rendering we do not need for a private ops console. Flutter would split the team. Vue has no existing skill in this group. Create React App is obsolete.
- **Benefit:** Fastest credible path to a GIS dashboard the team can style consistently (safe/assist/danger colours).
- **Limitation:** Not a native mesh client; offline is a PWA cache.
- **Connection:** Talks REST to FastAPI; Leaflet owns the map; TanStack Query owns polling.

### 11.3 React Router and TanStack Query

Classification: Frontend

- **Definition:** Client URL routing, and a library for fetch/cache/poll of API data.
- **Why AASHRAY uses it:** Judges must open `/ops?tick=7&incident=A2719`. Snapshot changes every demo tick without Redux ceremony.
- **Where it is used:** Role trees; useQuery on `/ops/snapshot` every ~2s; mutations for SOS, check-in, assign, demo/play.
- **Why not the alternative:** A useState “screen machine” cannot bookmark a disaster tick. Redux is extra for one document. Context holds role + clock id only.
- **Benefit:** Shareable demo states; fewer stale-UI bugs.
- **Limitation:** Polling uses a little bandwidth; 2–3s is acceptable.
- **Connection:** Snapshot JSON is the contract with the backend.

### 11.4 Leaflet + GeoJSON (not Google Maps)

Classification: GIS (client)

- **Definition:** Open-source 2D map. Layers are GeoJSON FeatureCollections.
- **Why AASHRAY uses it:** Every AASHRAY decision is spatial. The map is the product.
- **Where it is used:** Ops: all layers with provenance popups. Citizen: own pin, zones, safest route, shelters — never others’ SOS.



- **Why not the alternative:** Google Maps / Mapbox GL: keys, cost, and the “wrapper around Maps” critique. OpenLayers is heavier. deck.gl is overkill for ~10 layers in one bbox. 3D globes do not add fusion.
- **Benefit:** OSM-native; GeoJSON in equals GeoJSON on screen; no tile vendor lock-in.
- **Limitation:** Weak photorealistic 3D terrain; optional static hillshade later.
- **Connection:** Backend returns features; client does not invent geometry except device GPS.

### 11.5 FastAPI (one Python process)

Classification: Backend

- **Definition:** Typed Python HTTP API. Pydantic models are the contract.
- **Why AASHRAY uses it:** Shapely, networkx, optional embeddings and HTTP must share an evidence row in the same request. Python is the language where that is natural.
- **Where it is used:** Ingest, fusion, decision engine, demo clock, snapshot, health.
- **Why not the alternative:** A second OCR/CV microservice (common in other student projects) dies on two ports and jury Wi-Fi. Express/Nest makes GIS weaker. Django is slower to iterate. Go has no in-team GIS practice. Flask is less structured.
- **Benefit:** One uvicorn command runs the intelligence loop.
- **Limitation:** In-process CPU is enough for tens of evidence rows, not a national ingest cluster.
- **Connection:** SQLAlchemy; optional httpx to LLM/OSRM; reads /scenario files.

### 11.6 SQLite + SQLAlchemy 2 + Alembic (not PostGIS in MVP)

Classification: Database

- **Definition:** File database, ORM, versioned migrations. Geometry stored as GeoJSON text; spatial predicates in Shapely.
- **Why AASHRAY uses it:** One district bbox, well under a thousand rows, must run from a laptop.
- **Where it is used:** users, evidence, incidents, membership, check-ins, blocked\_edges, alert\_events, demo\_state.
- **Why not the alternative:** PostGIS is the correct scale-up, not the prototype (Docker/ops cost). Mongo weakens evidence↔incident relations. Firebase RTDB is the wrong query model. Spatialite is poorly documented for this team.
- **Benefit:** The whole demo is a file. Matches student lab reality.
- **Limitation:** Not for high concurrent state-scale writers.
- **Connection:** API session per request; Pydantic validates GeoJSON before persist.

### 11.7 PyJWT + bcrypt (not Firebase for this product)

Classification: Security / database

- **Definition:** Hashed passwords; signed tokens carrying user id and role (citizen | responder).
- **Why AASHRAY uses it:** Exact coordinates and victim photos are responder-only. Roles are a privacy control, not a login fashion.
- **Where it is used:** Seeded citizen@demo and ops@demo. Media and snapshot extras require responder role.
- **Why not the alternative:** Hosted identity (Firebase/Auth0) adds console setup and token expiry mid-pitch. Anonymous SOS with no identity is a hoax magnet; we still rate-limit.
- **Benefit:** Deterministic login on a projector.
- **Limitation:** Not production IAM. Future: agency SSO. JWT\_SECRET must not remain a default.
- **Connection:** Every evidence row has user\_id.

### 11.8 SlowAPI

Classification: Deterministic rules

- **Definition:** In-process HTTP rate limiter.
- **Why AASHRAY uses it:** False SOS starts at ingest.
- **Where it is used:** Stricter quota on POST /evidence than on GET /health.

- **Why not the alternative:** Cloud WAF is overkill for a prototype.
- **Benefit:** Simple abuse brake.
- **Limitation:** Does not stop many-account coordination (out of MVP threat model).
- **Connection:** Sits in front of fusion.

### 11.9 Shapely + committed scenario GeoJSON

Classification: GIS + simulated scenario data

- **Definition:** Shapely: point-in-polygon, buffer, line–polygon intersection. Scenario pack: rain frames, slope mask, settlements, roads, POIs, visibility grid, valley clip, candidate routes.
- **Why AASHRAY uses it:** The pitch must be reproducible. Live national WMS and live Overpass are not a student SLA. Circles in Leaflet would make this a generic SOS app.
- **Where it is used:** Decision engine after each fusion or tick; Leaflet displays copies in snapshot layers.
- **Why not the alternative:** Turf.js only on the client would let backend and UI disagree. GeoPandas is optional weight if Shapely suffices.
- **Benefit:** One spatial truth for alerts and for route rejection.
- **Limitation:** Buffers are geometric, not hydrological flow. Rain/risk files are simulated unless labelled otherwise.
- **Connection:** Feeds zones, PIP, route scoring.

### 11.10 OpenStreetMap tiles and offline Overpass extract

Classification: GIS; Overpass is prep-time external API

- **Definition:** OSM: volunteer global map. Tiles: basemap. Overpass: query API used once while authoring, then saved.
- **Why AASHRAY uses it:** Hospitals, fire stations, roads and settlements must be real enough for resource matching and isolation.
- **Where it is used:** Leaflet tile layer (fair use). POIs/roads committed in /scenario — not queried on stage.
- **Why not the alternative:** Google Places: cost and ToS. Live Overpass during demo: timeouts. Paid tile APIs: keys.
- **Benefit:** Credible roads offline after extract.
- **Limitation:** OSM completeness in hills varies; the bbox must be verified by the team.
- **Connection:** Feeds networkx and resource ranker.

### 11.11 networkx (access-cut)

Classification: GIS + deterministic rules

- **Definition:** Graph library. Nodes = junctions/settlements; edges = simplified ways with blocked flags. Isolation = BFS reachability from a hub after removing blocked edges.
- **Why AASHRAY uses it:** A landslide 200 m from a village is not the same as cutting the only road. Pin proximity cannot express that.
- **Where it is used:** On blocked\_edges insert: BFS; unreachable settlements get isolated=true and a one-hop cascade (medical access delayed).
- **Why not the alternative:** Asking an LLM “is the village cut off?” is not evidence. pgRouting/custom C++ routers are weeks of work.
- **Benefit:** Explainable graph fact: “edge 14 is the only access.”
- **Limitation:** Simplified topology, not every footpath. Enough for one valley.
- **Connection:** Writes incident.isolated; multiplies priority; citizen copy that hospital is unreachable by road.

### 11.12 Precomputed route polylines + optional OSRM

Classification: GIS + optional external API + precomputed fallback

- **Definition:** OSRM is an open driving router on OSM. AASHRAY stores 2–3 candidate LineStrings per origin–destination (OSRM-shaped). Live OSRM is used only if configured and healthy.

- **Why AASHRAY uses it:** AASHRAY ranks safety. The pitch must not die because a public demo router is down. The team also must not write a router.
- **Where it is used:** GET /routes/safest: intersect hazard/blocked → reject or penalize; else min-risk then duration. UI: SAFEST AVAILABLE vs REJECTED (reason).
- **Why not the alternative:** Google Directions as the product thesis is a Maps clone. Leaflet Routing Machine with no fallback outages. Mapbox as the only path: key + outage. Self-hosted Valhalla/OSRM on a full PBF is too heavy for MVP (future).
- **Benefit:** Honest safety-over-ETA that works offline.
- **Limitation:** Not a national multimodal engine.
- **Connection:** health.routing = live | precomputed.

### 11.13 LLM (Groq or Gemini Flash) — thin, optional

Classification: AI + external API, non-blocking

- **Definition:** Hosted large language model for short completions.
- **Why AASHRAY uses it:** Citizen text is messy. We need structured fields and a 20-word sitrep from JSON the engine already computed.
- **Where it is used:** classify(raw\_text); sitrep(snapshot\_slice). Timeout ~4 seconds.
- **Why not the alternative:** Local Llama on a demo laptop is slow. Fine-tuning has no dataset. Using an LLM for routing, isolation or priority is indefensible.
- **Benefit:** Language help without training. health.llm=down is a valid, rehearsed state.
- **Limitation:** Hallucination if allowed to invent geometry — therefore it never writes coordinates, blocked edges or ranks.
- **Connection:** httpx client; fusion hot path does not import it.

### 11.14 Embeddings with TF-IDF fallback

Classification: AI with deterministic fallback

- **Definition:** Embeddings: dense vectors so paraphrases are close in cosine similarity. TF-IDF: classic sparse similarity in scikit-learn.
- **Why AASHRAY uses it:** Fusion's hard case is different words for one event ("landslide" / "road blocked" / "trapped" / debris caption).
- **Where it is used:** text\_sim term in the fusion score. API vectors when present; else TfidfVectorizer on the small corpus.
- **Why not the alternative:** Fine-tuned BERT needs GPU and labels. Shipping sentence-transformers as the only path means a large download on the jury laptop. LLM "are these the same?" on every pair is slow and unstable.
- **Benefit:** Semantic link with a boring offline fallback.
- **Limitation:** TF-IDF is weaker on paraphrase; API has latency.
- **Connection:** Fusion only — not routing or visibility.

### 11.15 Optional vision

Classification: Optional AI

- **Definition:** Image-to-tags (debris, water, obstruction) with low confidence.
- **Why AASHRAY uses it:** Photo evidence should not be unstructured forever — but it must not gate SOS.
- **Where it is used:** Async tags on the locker. If skipped, a labelled scripted tag on the demo photo is more honest than a flaky call.
- **Why not the alternative:** Training a CNN is out of scope. Packaging-OCR pipelines from other projects are the wrong domain.
- **Benefit:** Structured photo evidence without overclaiming CV.
- **Limitation:** APIs fail; accuracy is not field-validated.
- **Connection:** Never changes geometry by itself.

### 11.16 PWA cache + IndexedDB SOS queue

Classification: Client + database when flushed

- **Definition:** Installable web app with last-snapshot cache and a client queue.
- **Why AASHRAY uses it:** Networks die in the disasters we claim to care about. Mesh radio is not demoable here.
- **Where it is used:** Cache shelters, last safest route, last zone. Flush SOS when POST works. Show an SMS-shaped card (no DLT send in MVP).
- **Why not the alternative:** Native Android + BLE mesh: weeks and unverifiable in three minutes.
- **Benefit:** Honest connectivity story.
- **Limitation:** No offline recomputation of new routes; last known only.
- **Connection:** Queue drains into POST /evidence.

### 11.17 Recharts, pytest, Playwright, local hosting

Classification: Quality / ops

- **Definition:** Charts already known to the team; unit tests; browser e2e; uvicorn+vite on a laptop.
- **Why AASHRAY uses it:** Priority and visibility must be visible numbers. Fusion/isolation must be testable without an LLM. The pitch must not depend on Kubernetes.
- **Where it is used:** CI and rehearsal. Optional later: Render/Railway API + static frontend.
- **Why not the alternative:** D3-from-scratch wastes a person. K8s/Terraform/ECS are anti-hackathon. Vercel Python serverless fights Shapely wheels.
- **Benefit:** Rehearsable, inspectable, portable.
- **Limitation:** Local-first does not prove public-scale hosting — and we do not claim it.
- **Connection:** Tests use scenario fixtures ticks 2–7 with LLM down.

## 12. Data model and APIs

| Table             | Role                                                                         |
|-------------------|------------------------------------------------------------------------------|
| users             | role citizen responder; last_lat/lon; password hash                          |
| evidence          | atomic signal + structured fields + optional embedding + source + media_path |
| incidents         | centroid, geometry, confidence, status, isolated, sitrep cache               |
| incident_evidence | membership and contribution_score                                            |
| checkins          | safe   assist   danger                                                       |
| blocked_edges     | road closures                                                                |
| alert_events      | anti-spam: one row per user/incident/level                                   |
| demo_state        | tick, rainfall_index, playing                                                |

Table 6. Evidence is not an incident. That split is the product.

| Endpoint                                | Job                                  |
|-----------------------------------------|--------------------------------------|
| POST /auth/login                        | JWT with role                        |
| POST /evidence                          | Create signal; run fusion + decision |
| GET /incidents/{id}                     | Locker + score breakdown             |
| GET /ops/snapshot                       | One document for the dashboard       |
| POST /ops/incidents/{id}/assign dismiss | Simulated response action            |
| GET /routes/safest                      | Score candidates                     |
| POST /demo/tick/demo/play               | Scenario clock                       |

| Endpoint        | Job                                       |
|-----------------|-------------------------------------------|
| GET /health     | db; llm ok down; routing live precomputed |
| GET /media/{id} | Responder-only photo                      |

Table 7. Citizens get 201 on SOS after persist; fusion is in-process because volume is tiny.

## 13. Incident fusion, AI, and decision engine

### Fusion (complete)

On ingest: persist; optional classify; find candidate incidents within about 500 m and 45 minutes with compatible category; score; attach if  $\text{max} \geq 0.65$  else new incident; store members; run decision engine. Responders may dismiss or split. Conflicting categories set a disagreement flag rather than silently averaging.

```
score = 0.40*geo(haversine) + 0.25*time_decay + 0.20*category_match + 0.15*text_sim
attach if score >= 0.65 else new incident
confidence = score # UI label: cluster confidence, NOT field accuracy
```

### Decision engine (AI does not sit here)

| Module                          | Output                            | Class                               |
|---------------------------------|-----------------------------------|-------------------------------------|
| Hazard restyle from tick        | Risk polygons labelled estimation | Simulated + GIS                     |
| Buffer $\cap$ valley mask       | Critical / Warning / Nearby       | GIS + rules                         |
| Point-in-polygon + alert_events | Who is notified                   | GIS + rules                         |
| Corroboration                   | Critical notify true/false        | Rules                               |
| Visibility grid                 | 0–1 + uncertain copy              | Rules + simulated denominator       |
| Priority weights on screen      | Ranked incidents                  | Rules                               |
| POI rank by type and access     | Suggested resource                | GIS + rules; availability simulated |
| BFS after blocked last edge     | isolated + medical-delay          | Rules + GIS                         |
| Polyline $\cap$ hazard          | Safest vs rejected                | GIS + rules                         |

Table 8. If asked “where is the AI in priority?” the answer is: nowhere, by design.

Illustrative priority (constants shown in UI, tunable):  $3 \cdot \text{severity} + 0.15 \cdot \text{people} + 2 \cdot \text{critical\_checkins} + 0.4 \cdot \text{hours} - 0.05 \cdot \text{km}$ , then an isolation multiplier. A juror must be able to recompute the rank by hand. This is not a learned model.

## 14. GIS, routing, isolation, alerts, visibility

**Geofence:** not radius-N. Buffer fused geometry, intersect a pre-authored valley/slope mask so Critical follows downslope and the blocked road. Nearby means do not travel toward the incident. **Visibility:** committed grid; denominator = seeded demo users; low visibility + nearby hazard  $\rightarrow$  information-gathering, not “people missing.” **Resources:** type match then graph access, not crow-flies through a cliff. **Routing:** safety first among N candidates. **Isolation:** graph theorem, then one-hop cascade text — not a separate deep-learning “disaster cascade engine.”

## 15. Complete data flow and tick-by-tick workflow

*Citizen  $\rightarrow$  Evidence  $\rightarrow$  Backend  $\rightarrow$  AI/GIS/Rules  $\rightarrow$  Fused Incident  $\rightarrow$  Decision Engine  $\rightarrow$  Responder  $\rightarrow$  Action  $\rightarrow$  Verification  $\rightarrow$  loop*

- 1 Capture PWA or demo clock: SOS/report/photo-meta/check-in (source=live|simulated)
- 2 Persist Validate, rate-limit, write evidence
- 3 Interpret Optional LLM fields; keyword fallback; optional vision tags
- 4 Fuse geo + time + category + text\_sim  $\rightarrow$  incident + confidence
- 5 Decide zones, PIP alerts, visibility, blockage, BFS isolation,

|           |                                                 |
|-----------|-------------------------------------------------|
|           | priority, resources, safest-of-N                |
| 6 Picture | GET /ops/snapshot                               |
| 7 Act     | assign/dismiss; citizen sees zone/status        |
| 8 Verify  | new check-ins; more evidence; next tick; repeat |

Software mapping of Table 5: T0 seed snapshot; T1 rainfall\_index++; T2 four evidence rows; T3 attach into one cluster; T4 zone polygons; T5 check-in driven visibility; T6 route scoring; T7 blocked\_edges += last access; T8 assign + optional sitrep. The clock injects inputs; it does not bypass fusion or BFS.

## 16. Privacy, misinformation, and failure modes

- Citizen map never shows others' exact SOS or photos. Photos live under MEDIA\_DIR; GET is responder-JWT only.
- Retention for the hackathon is the demo session. Do not keep victim images after the pitch if avoidable.
- CORS locked to the frontend origin when deployed.
- LLM/OSRM/vision timeouts fall back; snapshot still returns last good sitrep; decision engine never imports the LLM client.
- Demo tick is an integer so a crashed laptop can resume.
- Misinformation: source count, disagreement flag, human dismiss — fusion must not boost the first viral sentence.

## 17. Real versus simulated data

| Real in the prototype                                | Simulated (must label source=simulated)              |
|------------------------------------------------------|------------------------------------------------------|
| Fusion, priority, isolation, PIP, route/polygon math | Rainfall / river / IMD unless actually wired         |
| Cached OSM roads and POIs (extract)                  | Registered population (e.g. 3000)                    |
| Optional live GPS from team phones                   | Ambulance availability toggles                       |
| Optional LLM classify/sitrep if key present          | Extra citizen reports injected by the clock          |
| Optional live OSRM if healthy                        | Network-degraded flag; precomputed routes as default |

Table 9. A layer without provenance is a bug. Honesty is part of the architecture.

## 18. Team roles and implementation plan

**Person 1 — Frontend and UX** (citizen + ops, Leaflet, copy, PWA, Playwright). **Person 2 — Backend and database** (schema, JWT, ingest, snapshot, demo clock, API tests). **Person 3 — AI and GIS/data** (scenario pack, Shapely, networkx, fusion score, optional LLM, route scoring). Person 3 owns the fusion formula; Person 2 owns the tables. Person 1 does not invent GeoJSON the backend does not know (except device GPS). Nobody trains a custom landslide net or stands up Kubernetes.

| Order | Phase                                  | Done when                                        |
|-------|----------------------------------------|--------------------------------------------------|
| 1     | Greenfield aashray/                    | Empty apps run; env documented                   |
| 2     | Scenario pack                          | Map shows roads, settlements, POIs, rain 0, grid |
| 3     | Schema + JWT + snapshot stub           | Login and tick 0                                 |
| 4     | Citizen evidence + check-ins           | SOS row + pin                                    |
| 5     | Rules fusion + locker                  | Four fixtures → one incident, no LLM             |
| 6     | Zones, visibility, priority, resources | Weights and non-circle zones                     |
| 7     | Isolation BFS + safest-of-N            | T7 wow with LLM down                             |
| 8     | Corroboration + alert_events           | Single SOS does not Critical-spam                |
| 9     | Optional LLM/embeddings/vision flags   | Both llm down and ok rehearsed                   |

| Order | Phase                   | Done when                                    |
|-------|-------------------------|----------------------------------------------|
| 10    | E2E + provenance labels | Three-minute clock; simulated badges visible |

Table 10. LLM last. Scenario, schema and map first.

## 19. Evaluation criteria and judge questions

### How this work should be judged

- **Thesis fidelity:** is this still an SOS app, or do four signals become one incident with a locker?
- **Honesty:** are rain, population and availability labelled? Is unconfirmed not missing?
- **GIS that decides:** does a fast route get rejected? Does last-road blockage isolate a polygon?
- **Explainability:** can a human recompute priority and fusion?
- **Resilience:** does T3–T7 work with health.llm=down?
- **Scope discipline:** one scenario deep, not ten hazards shallow.
- **Social care:** corroboration, anti-spam, photo gating, no fake government dispatch.

### Anticipated technical questions

| Question             | Defensible answer                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| Just SOS + a map?    | SOS is evidence. Operational object is a fused incident, then graph isolation and route/hazard.           |
| Why not PostGIS?     | One bbox. Shapely is the right complexity. PostGIS is the scale-up.                                       |
| LLM down?            | Keyword + TF-IDF fusion. Isolation and priority never called the LLM.                                     |
| 91% accurate?        | Cluster score, not a validated detector. UI must say so.                                                  |
| 658 missing?         | Simulated registered users; unconfirmed_safety_status.                                                    |
| Alert NDMA?          | No. Simulated ops console only.                                                                           |
| Custom router?       | We rank candidate geometries. Shortest path is OSRM's solved job.                                         |
| False SOS?           | Rate limit, unverified single source, corroboration, dismiss.                                             |
| Live weather?        | Labelled simulated unless wired.                                                                          |
| Scale to a state?    | Not on SQLite. Algorithms transfer; runtime becomes PostGIS + queued fusion + district extracts + IAM.    |
| Where is the AI?     | Language and optional paraphrase-linking. Geography and ranking are deterministic so they can be audited. |
| Why not Google Maps? | OSM is the road/POI source; Google would make us a Maps wrapper.                                          |
| Why one process?     | Fusion must share a row with Shapely. A second model server is a demo failure mode.                       |

Table 11. Evaluation FAQ.

## 20. Limitations, scalability, and future work

This prototype is a hill-district laboratory, not a certified warning system. Geometric buffers are not catchment hydrology. OSM can be incomplete. Fusion can over-merge or under-merge; humans can split/dismiss. Visibility denominators are seeded users. SQLite will not take a state-scale write load. Scale path: PostgreSQL/PostGIS, queued fusion, per-district OSM/OSRM, real identity, optional live weather with fallback, household check-in, official agency roles, production SMS under TRAI DLT. Those are sequential upgrades to the same thesis, not a reason to start with Kubernetes.

## 21. Conclusion

AASHRAY's research claim is modest and testable: in a time-evolving landslide scenario, fragmented citizen signals can be fused into one explainable incident; safety silence can be mapped as visibility rather than as missing



persons; a fast unsafe path can be rejected; and a last remaining road can be shown to isolate a settlement — all with provenance labels, and all without a language model being allowed to move the map. That is a disaster intelligence prototype. It is not “SOS with AI.”

*Build rule: if two designs conflict, prefer the one that still works with DEMO\_MODE=1, no LLM key, no OSRM, and a laptop that has already loaded the scenario pack.*

## Appendix A. Final stack table

| Layer        | Technology                        | Purpose              | Why chosen              |
|--------------|-----------------------------------|----------------------|-------------------------|
| UI           | React 19 + Vite + TS + Tailwind 4 | Citizen + ops SPAs   | Fast, known, sufficient |
| App routing  | React Router                      | Deep links           | Demo ticks must be URLs |
| Server state | TanStack Query                    | Snapshot poll        | No Redux                |
| Map          | Leaflet + GeoJSON                 | Decision layers      | OSM-native, free        |
| Charts       | Recharts                          | Weights / visibility | Known library           |
| Offline      | PWA + IndexedDB                   | Cache + SOS queue    | No mesh                 |
| API          | FastAPI one process               | Ingest–decide–clock  | Python GIS+AI together  |
| DB           | SQLite + SQLAlchemy + Alembic     | Evidence/incidents   | Portable demo           |
| Auth         | PyJWT + bcrypt                    | Two roles            | No identity vendor      |
| Spatial      | Shapely + scenario files          | PIP, ∩, buffer       | PostGIS later           |
| Graph        | networkx                          | Isolation            | Auditable BFS           |
| Roads/POIs   | OSM extract                       | Resources + graph    | Overpass offline        |
| Paths        | Precomputed; optional OSRM        | Safest-of-N          | Demo-safe               |
| AI text      | Groq or Gemini Flash              | Classify + sitrep    | Thin, optional          |
| Embeddings   | API + TF-IDF                      | Paraphrase fusion    | Works offline           |
| Vision       | Optional / scripted tags          | Photo evidence       | Must not gate SOS       |
| Rate limit   | SlowAPI                           | False SOS            | Simple                  |
| Test         | pytest + Playwright               | Loop proof           | LLM-down CI             |
| Host         | Laptop first                      | Pitch SLA            | Not K8s                 |

Table 12. Rejected: Firebase-as-core, PostGIS-in-MVP, Kafka, Kubernetes, custom DL, live Overpass on stage, Google Maps as thesis, WebSocket-only UI, mesh, fake NDMA SMS.

## Appendix B. Claims we will not make

- We do not issue official public warnings or replace 112, police, fire, NDRF, SDRF or district authorities.
- We do not treat unconfirmed app users as missing persons.
- We do not present scenario rainfall or authored hazard polygons as live IMD/CWC science unless they are.
- We do not present clustering confidence as peer-reviewed accuracy.
- We do not put identifiable victim photographs on a public map.
- We do not claim lives saved. Demo metrics are time-to-corroborated-picture and visibility coverage of the scenario area.

Document: AASHRAY Full Research and Technical Design. Viksit Bharat Young Leaders Dialogue — Hack for Social Cause. September 2026. Three-person student team. Prototype research for evaluation — not operational doctrine.